

Leyang Li

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RESEARCH INTERESTS

Human-AI Interaction, Human-Centered AI, Interactive Systems, LLM Social Simulation, Usable Privacy and Security

EDUCATION

Johns Hopkins University

Ph.D. in Computer Science, Advised by Prof. Yaxing Yao

Aug. 2026–Present

University of Notre Dame

B.S. in Computer Science, Advised by Prof. Toby Jia-Jun Li

Aug. 2022–May 2026

GPA: 3.95/4.00

SELECTED PUBLICATIONS AND PREPRINTS

(* Denotes equal contribution)

- [C.2] **AROMA: Mixed-Initiative AI Assistance for Non-Visual Cooking by Grounding Multimodal Information Between Reality and Videos** 📄 📺
Zheng Ning, **Leyang Li**, Daniel Killough, JooYoung Seo, Patrick Carrington, Yapeng Tian, Yuhang Zhao, Franklin Mingzhe Li, Toby Jia-Jun Li
In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST'25)
- [C.1] **Why am I seeing this: Democratizing End User Auditing for Online Content Recommendations** 📄 📺
Chaoran Chen, **Leyang Li**, Luke Cao, Yanfang Ye, Tianshi Li, Yaxing Yao, Toby Jia-Jun Li
In Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST'25)

RESEARCH EXPERIENCE

SaNDwich Lab | University of Notre Dame

Advisor: Prof. Toby Jia-Jun Li

Sept. 2023–May 2026

Notre Dame, IN

- **Non-Visual Cooking:** A mixed-initiative assistant that grounds information between real-world cooking and recipe videos to support blind/low-vision cooks [C.2].
 - * Motivation: The highly visual nature of recipe videos creates a barrier for blind and low-vision (BLV) cooks, disconnecting them from their non-visual sensory experience of cooking.
 - * Contribution: Introduced a mixed-initiative AI assistant that grounds video recipe instructions with the user's real-world cooking context. The system integrates the user's non-visual feedback with a wearable camera to provide both reactive and proactive assistance.
 - * Result: A user study with BLV participants demonstrated AROMA's high usability and effectiveness, yielding design implications for accessible, context-aware AI assistants that leverage users' perceptual abilities.
- **Privacy Auditing Sandbox:** An interactive interface that democratizes end-user auditing of online content recommendations [C.1].
 - * Motivation: Opaque recommendation systems disempower users and erode trust by preventing them from understanding how their data is used.
 - * Contribution: Developed an interactive sandbox that allows users to audit recommendation systems by creating and controlling AI-generated virtual personas. The system simulates how algorithms respond to these personas and uses LLMs to analyze and visualize the causal links between user attributes and the content they are shown.
 - * Result: User studies showed the sandbox improved users' understanding of algorithms and revealed opaque recommendation factors, informing guidelines for usable transparency and value-aligned AI.

SPUD Lab | Carnegie Mellon University

Advisor: Prof. Sauvik Das

Jun. 2025–Feb. 2026

Pittsburgh, PA

- **SP-ABC Bench:** A benchmark to simulate and evaluate LLM agents' privacy attitude and behavior [P.1].
 - * Motivation: Practitioners lack scalable, user-aligned methods to anticipate population-level security and privacy behavior when designing new products.
 - * Contribution: Constructed a 30-task benchmark that evaluates LLM simulation alignment with S&P behavior, attitude, and reasoning, and applied it to 12 LLMs, 4 persona strategies, and a theory-grounded prompt.
 - * Result: Larger or newer LLMs do not consistently improve alignment; scenario-primed personas achieve the best performance on behavioral tasks; and a Privacy Calculus- and Bounded Rationality-grounded prompt improves overall simulation quality.

SELECTED PROJECTS

SafePrompt, Winner project (2nd place) at 2026 Hesburgh Libraries Hackathon A middle layer between users and AI chatbots to reduce private information disclosure.	2026 
MindEcho, Winner project (1st place) at 2025 Hesburgh Libraries Hackathon A meeting assistant with real-time transcription, mind map visualization, and keyword-based auto-completion.	2025 
P2P DHT, Course project of CSE 40771 Distributed Systems A peer-to-peer decentralized hash table with crash detection and recovery based on Chord, chain replication, and RPC.	2024
A11yVate, Winner project (2nd place) at 2024 Hesburgh Libraries Hackathon A crowdsourcing campus accessibility map platform with path optimization and customized LLM-suggestions.	2024

TEACHING EXPERIENCE

Teaching Assistant, CSE 40771 Distributed Systems Instructor: Prof. Douglas Thain	Spring 2026 CSE Department, University of Notre Dame
ACES Tutor, CSE 30341 Operating System Principles Instructors: Prof. Peter Bui (Fall 25'), Prof. Fanxin Kong (Spring 26')	Fall 2025, Spring 2026 College of Engineering, University of Notre Dame
Teaching Assistant, CSE 30341 Operating System Principles Instructor: Prof. Douglas Thain	Spring 2025 CSE Department, University of Notre Dame

INDUSTRIAL EXPERIENCE

Software Engineer Intern China Construction Bank Fintech Big Data System Maintenance (Flink-based Kafka consumer)	Jun.–Aug. 2024 Shanghai, China
Data Analyst Intern Everbright Securities Asset Management CO., LTD Machine Learning for Investment (GRU with multi-head self-attention)	Jul.–Aug. 2023 Shanghai, China

SELECTED HONORS & AWARDS

Hesburgh Libraries Hackathon 2026 - 2 nd Place (\$2,000)	2026
Meruelo Family Summer Research Funding (\$4,500)	2025
Hesburgh Libraries Hackathon 2025 – 1 st Place (\$3,000)	2025
American Statistical Association Data Fest - Best Use of External Data (\$1,000)	2025
Frank Reilly Scholar (\$5,000)	2024
Hesburgh Libraries Hackathon 2024 - 2 nd Place (\$2,000)	2024
Dean's Honor List, University of Notre Dame	All Semesters 2023–2025
Greater China Scholar (\$5,000)	2022

SKILLS

Programming: Python, JavaScript & TypeScript, C/C++, Java, SQL, R, and others
Tools: React, Flask, Sanic, Django, PyTorch, sklearn, Figma, LaTeX, and others
Other: Photography, Procreate, Sketching, Gouache, Oil Painting
Languages: English, Mandarin